Linear Algebra

Week 5

Contents

1 Vector spaces 2

2 Vector subspaces 5

3 Intersections, unions, and sums of subspaces 8

4 Linear combinations and span 11

1 Vector spaces

Linear algebra studies structures in which we can add objects and multiply them by scalars. The objects may be familiar coordinate vectors, but they may also be matrices, polynomials, or functions. We call them *vectors* because they obey the same algebraic rules.

One reason for isolating these rules is that they let us discuss linear maps. A linear map preserves the two operations:

$$T(u+v) = T(u) + T(v), \qquad T(\alpha v) = \alpha T(v).$$

To state these conditions, we do not need a product of two vectors. Vector addition and scalar multiplication are enough.

1.1 The definition and its ten conditions

Definition 1.1 (Vector space over a field)**.** Let $\mathbb{F}$ be a field and V a set equipped with vector addition and scalar multiplication. We say that V is a *vector space over* $\mathbb{F}$ if the following conditions hold.

1. **Closure under addition.** For every $u, v \in V$, their sum $u + v$ is defined and belongs to V.
2. **Associativity of addition.** For every $u, v, w \in V$,
$$(u+v)+w = u+(v+w).$$
3. **Commutativity of addition.** For every $u, v \in V$, $u + v = v + u$.
4. **A zero vector.** There is an element $\mathbf{0} \in V$ such that $v + \mathbf{0} = v$ for every $v \in V$.
5. **Additive inverses.** For every $v \in V$, there is $w \in V$ such that $v + w = \mathbf{0}$.
6. **Closure under scalar multiplication.** For every $\alpha \in \mathbb{F}$ and $v \in V$, the scalar multiple αv is defined and belongs to V.
7. **The scalar identity acts as an identity.** For every $v \in V$, $1v = v$.
8. **Distributivity over vector addition.** For every $\alpha \in \mathbb{F}$ and $u, v \in V$,
$$\alpha(u+v) = \alpha u + \alpha v.$$
9. **Distributivity over scalar addition.** For every $\alpha, \beta \in \mathbb{F}$ and $v \in V$,
$$(\alpha+\beta)v = \alpha v + \beta v.$$
10. **Compatibility of scalar multiplication.** For every $\alpha, \beta \in \mathbb{F}$ and $v \in V$,
$$(\alpha \times \beta)v = \alpha(\beta v).$$

The product $\alpha \times \beta$ is multiplication in the field. Scalar multiplication is a map $\mathbb{F} \times V \to V$: it takes one scalar and one vector as inputs, and its output is a vector. We may write its value as $\alpha \times v$, or simply αv without an explicit multiplication symbol. No multiplication of two vectors is required by the definition.

Remark 1.2 (The zero scalar and the zero vector)**.** The symbol $0 \in \mathbb{F}$ is a scalar, while $\mathbf{0} \in V$ is a vector. They have different roles. In familiar spaces, the zero vector may be a zero column, a zero matrix, the zero polynomial, or the zero function. When the meaning is clear, both are often written simply as 0.

Example 1.1 (The familiar real plane)**.** The set

$$\mathbb{R}^2 = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} : x, y \in \mathbb{R} \right\}$$

with coordinatewise addition and scalar multiplication is a real vector space. For example,

$$\begin{pmatrix} 2 \\ -1 \end{pmatrix} + \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}, \qquad 2 \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}.$$

The zero vector is $(0, 0)^\mathsf{T}$, and the additive inverse of $(x, y)^\mathsf{T}$ is $(-x, -y)^\mathsf{T}$.

1.2 Standard vector spaces and their notation

Definition 1.3. Let $\mathbb{F}$ be a field.

1. $\mathbb{F}^n$ is the set of all column vectors of length n with entries in $\mathbb{F}$.
2. $M_{m,n}(\mathbb{F})$ is the set of all $m \times n$ matrices over $\mathbb{F}$. The notation $\mathbb{F}^{m \times n}$ is also used.
3. $M_n(\mathbb{F}) = M_{n,n}(\mathbb{F})$ is the set of square matrices of order n.
4. $\mathbb{F}_n[x]$ is the set of all polynomials over $\mathbb{F}$ of degree at most n, together with the zero polynomial.
5. $\mathbb{F}[x]$ is the set of all polynomials over $\mathbb{F}$, with no fixed degree bound.

Theorem 1.4 (Standard examples)**.** With their usual addition and scalar multiplication,

$$\mathbb{F}^n, \qquad M_{m,n}(\mathbb{F}), \qquad \mathbb{F}_n[x], \qquad \mathbb{F}[x]$$

are vector spaces over $\mathbb{F}$.

Proof. For vectors and matrices, addition and scalar multiplication act on each entry. Every axiom follows by applying the corresponding field rule to each entry. For polynomials, apply the same reasoning to each coefficient. The sum of polynomials of degree at most n still has degree at most n, unless it is zero, which is explicitly included. Scalar multiplication also preserves this degree bound. □

Example 1.2 (A space of rectangular matrices)**.**

$$M_{2,3}(\mathbb{F}) = \left\{ \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} : a, b, c, d, e, f \in \mathbb{F} \right\}.$$

Its vectors are matrices. They can be added and multiplied by scalars, even though the product of two 2×3 matrices is not defined.

Example 1.3 (Exactly degree two does not work)**.** Consider

$$E = \{ax^2 + bx + c : a, b, c \in \mathbb{R},\ a \neq 0\}.$$

Both $2x^2 + x$ and $-2x^2 + 3$ belong to E, but their sum $x + 3$ does not. Also, the zero polynomial is missing. Thus E is not a vector space with the usual operations.

The same problem occurs for the set of polynomials of *exactly* degree n, for any fixed $n \geq 0$: it excludes zero and is not closed under multiplication by the zero scalar. This is why $\mathbb{F}_n[x]$ uses "degree at most n" and includes zero.

Example 1.4 (The field of scalars is part of the structure)**.** The complex numbers form a vector space over $\mathbb{R}$. Complex numbers can be added, and multiplication by a real scalar keeps us in $\mathbb{C}$.

More generally, every vector space over $\mathbb{C}$ can also be regarded as a vector space over $\mathbb{R}$ by restricting its allowed scalars to real numbers. For instance, $\mathbb{C}^2$ can be treated as either a complex or a real vector space.

The reverse direction is not automatic. With the usual scalar multiplication, $\mathbb{R}^2$ is not a vector space over $\mathbb{C}$: multiplying $(1,0)^\mathsf{T}$ by i gives $(i,0)^\mathsf{T}$, which does not belong to $\mathbb{R}^2$.

1.3 Consequences of the axioms

Theorem 1.5 (Basic identities in a vector space)**.** Let V be a vector space over $\mathbb{F}$.

1. The zero vector is unique.
2. Each $v \in V$ has a unique additive inverse, denoted $-v$.
3. For every $\alpha \in \mathbb{F}$ and $v \in V$, $\alpha\mathbf{0} = \mathbf{0}$ and $0v = \mathbf{0}$.
4. If $\alpha v = \mathbf{0}$, then $\alpha = 0$ or $v = \mathbf{0}$.
5. For every $\alpha \in \mathbb{F}$ and $v \in V$,
$$\alpha(-v) = (-\alpha)v = -(\alpha v).$$

Proof. If $\mathbf{0}$ and $\mathbf{0}'$ are both additive identities, then

$$\mathbf{0} = \mathbf{0} + \mathbf{0}' = \mathbf{0}'.$$

If w and z are both additive inverses of v, then

$$w = w + \mathbf{0} = w + (v + z) = (w + v) + z = \mathbf{0} + z = z.$$

Next, distributivity gives

$$\alpha\mathbf{0} = \alpha(\mathbf{0} + \mathbf{0}) = \alpha\mathbf{0} + \alpha\mathbf{0}.$$

Adding the inverse of $\alpha\mathbf{0}$ to both sides gives $\alpha\mathbf{0} = \mathbf{0}$. Similarly,

$$0v = (0 + 0)v = 0v + 0v$$

implies $0v = \mathbf{0}$.

If $\alpha v = \mathbf{0}$ and $\alpha \neq 0$, multiply by the inverse scalar:

$$v = 1v = (\alpha^{-1} \times \alpha)v = \alpha^{-1}(\alpha v) = \alpha^{-1}\mathbf{0} = \mathbf{0}.$$

Finally,

$$\alpha(-v) + \alpha v = \alpha((-v) + v) = \alpha\mathbf{0} = \mathbf{0},$$

so $\alpha(-v)$ is the inverse of αv. Also,

$$(-\alpha)v + \alpha v = ((-\alpha) + \alpha)v = 0v = \mathbf{0}.$$

Uniqueness of the inverse proves the last identities. □

1.4 The space of all functions

Let V be the set of all functions $f : \mathbb{R} \to \mathbb{R}$. Define addition and scalar multiplication pointwise:

$$(f + g)(x) = f(x) + g(x), \qquad (\alpha f)(x) = \alpha \times f(x).$$

These operations make V a real vector space. For example, for every $x \in \mathbb{R}$,

$$((f + g) + h)(x) = (f(x) + g(x)) + h(x) = f(x) + (g(x) + h(x)) = (f + (g + h))(x).$$

The other axioms are verified in the same way, using the real field rules at each input. The zero vector is the function that is zero at every input, and the inverse of f is the function $x \mapsto -f(x)$.

The identities just proved now apply automatically. For instance, if αf is the zero function, then $\alpha = 0$ or f is the zero function. This conclusion concerns the *whole function*, rather than merely one of its values.

2 Vector subspaces

2.1 The definition and the subspace test

Definition 2.1 (Vector subspace)**.** Let V be a vector space over $\mathbb{F}$, and let $W \subseteq V$. We call W a *vector subspace of* V if W, with the same addition and scalar multiplication as V, is itself a vector space over $\mathbb{F}$.

Example 2.1 (A plane through the origin and a shifted plane)**.** In $\mathbb{R}^3$, consider

$$U = \left\{ \begin{pmatrix} a \\ b \\ 0 \end{pmatrix} : a, b \in \mathbb{R} \right\}, \qquad H = \left\{ \begin{pmatrix} a \\ b \\ 3 \end{pmatrix} : a, b \in \mathbb{R} \right\}.$$

For U, sums and scalar multiples have the same required form:

$$\begin{pmatrix} a_1 \\ b_1 \\ 0 \end{pmatrix} + \begin{pmatrix} a_2 \\ b_2 \\ 0 \end{pmatrix} = \begin{pmatrix} a_1 + a_2 \\ b_1 + b_2 \\ 0 \end{pmatrix} \in U, \qquad \alpha \begin{pmatrix} a \\ b \\ 0 \end{pmatrix} = \begin{pmatrix} \alpha a \\ \alpha b \\ 0 \end{pmatrix} \in U.$$

The zero vector belongs to U, and the remaining axioms are inherited from $\mathbb{R}^3$. Thus U is a subspace.

In contrast, H does not contain the zero vector. It is also not closed under addition, since

$$\begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} + \begin{pmatrix} 2 \\ 4 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ 6 \end{pmatrix} \notin H.$$

The position of a plane relative to the origin matters.

Example 2.2 (A polynomial subspace and a finite-field non-example)**.** The set $\mathbb{C}_4[x]$ is a subspace of $\mathbb{C}[x]$: it contains zero and is closed under the usual operations.

Now take $V = M_{3,2}(\mathbb{Z}_5)$ and

$$E = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \\ 0 & c \end{pmatrix} : a, b, c \in \{1, 2, 4\} \right\}.$$

For instance, $\begin{pmatrix} 1 & 0 \\ 0 & 4 \\ 0 & 2 \end{pmatrix} \in E$, but $0_{3\times 2} \notin E$. Thus E is not a subspace. An inherited addition cannot acquire a different additive identity inside E; the subspace test below explains why.

Theorem 2.2 (Subspace test)**.** Let V be a vector space over $\mathbb{F}$ and $W \subseteq V$. Then W is a subspace if and only if all three conditions hold:

1. $W \neq \varnothing$.
2. For every $u, v \in W$, $u + v \in W$.

3. For every $\alpha \in \mathbb{F}$ and $u \in W$, $\alpha u \in W$.

When these conditions hold, the zero vector of W is the zero vector of V.

Proof. A subspace satisfies the conditions because it is a vector space with the inherited operations. Conversely, suppose the three conditions hold. Choose $w \in W$, which is possible by nonemptiness. Closure under scalar multiplication gives $0w = \mathbf{0} \in W$. It also gives $(-1)u = -u \in W$ for every $u \in W$. The addition and scalar multiplication are closed in W by assumption. All remaining axioms are equations already valid in V, so they hold for elements of W. Therefore W is a vector space. Since the ambient zero belongs to W, uniqueness of the additive identity excludes any different zero vector. □

Corollary 2.3 (A one-condition version)**.** A subset $W \subseteq V$ is a subspace if and only if $W \neq \varnothing$ and

$$\alpha u + v \in W \qquad \text{for all } \alpha \in \mathbb{F} \text{ and } u, v \in W.$$

Proof. A subspace has this property. Conversely, choose $w \in W$. Taking $u = v = w$ and $\alpha = -1$ gives $\mathbf{0} \in W$. Taking $v = \mathbf{0}$ gives closure under scalar multiplication, and taking $\alpha = 1$ gives closure under addition. Apply the subspace test. □

Remark 2.4. The sets $\{\mathbf{0}\}$ and V are always subspaces, called the *trivial subspaces*. When a set already lies in a known vector space, the subspace test usually avoids checking all ten axioms again.

2.2 Using the test on matrices and polynomials

Example 2.3 (A parametrized matrix subspace)**.** Define

$$U = \left\{ P(a,b) = \begin{pmatrix} 0 & a+2b \\ a-b & 2b \\ 3a & a+b \end{pmatrix} : a, b \in \mathbb{R} \right\} \subseteq M_{3,2}(\mathbb{R}).$$

Since $P(0,0) = 0_{3\times 2}$, the set is nonempty. Entrywise addition gives

$$P(a_1, b_1) + P(a_2, b_2) = P(a_1 + a_2, b_1 + b_2),$$

because, for example, the top-right entry is

$$(a_1 + 2b_1) + (a_2 + 2b_2) = (a_1 + a_2) + 2(b_1 + b_2).$$

Similarly, $\alpha P(a,b) = P(\alpha a, \alpha b)$. Thus U is a subspace.

Example 2.4 (Polynomials with a specified root)**.** Let

$$U = \{ f \in \mathbb{R}[x] : f(2) = 0 \}.$$

The zero polynomial belongs to U. If $f, g \in U$ and $\alpha \in \mathbb{R}$, then

$$(f+g)(2) = f(2) + g(2) = 0, \qquad (\alpha f)(2) = \alpha \times f(2) = 0.$$

Hence U is a subspace of $\mathbb{R}[x]$. Requiring $f(2) = 1$ instead would exclude the zero polynomial and would not define a subspace.

Example 2.5 (A complex matrix space)**.** Let

$$W = \left\{ Q(a,b) = \begin{pmatrix} a & b \\ -2b & 0 \\ 2a-b & a+4b \end{pmatrix} : a,b \in \mathbb{C} \right\}.$$

This is a subset of $M_{3,2}(\mathbb{C})$. Since $Q(0,0) = 0$, it is nonempty. Moreover,

$$Q(a_1,b_1) + Q(a_2,b_2) = Q(a_1+a_2, b_1+b_2), \qquad \alpha Q(a,b) = Q(\alpha a, \alpha b)$$

for every complex scalar α. Thus W is a complex vector space under the usual matrix operations. For example,

$$Q(1,-1) = \begin{pmatrix} 1 & -1 \\ 2 & 0 \\ 3 & -3 \end{pmatrix} \in W.$$

Example 2.6 (Symmetric matrices)**.** The set of symmetric matrices in $M_n(\mathbb{F})$ is a subspace over any field $\mathbb{F}$. Indeed, zero is symmetric, and if $A^{\mathsf{T}} = A$ and $B^{\mathsf{T}} = B$, then

$$(A+B)^{\mathsf{T}} = A^{\mathsf{T}} + B^{\mathsf{T}} = A + B, \qquad (\alpha A)^{\mathsf{T}} = \alpha A^{\mathsf{T}} = \alpha A.$$

The same calculation with a minus sign shows that skew-symmetric matrices also form a subspace. No restriction on $1+1$ is needed for either closure statement.

2.3 Solution sets of linear equations

Example 2.7 (A homogeneous system gives a subspace)**.** Solve over $\mathbb{R}$:

$$\begin{cases} x + 2y - z = 0, \\ 2x + 3y + z = 0. \end{cases}$$

Since the system is homogeneous, we can reduce its coefficient matrix:

$$\begin{pmatrix} 1 & 2 & -1 \\ 2 & 3 & 1 \end{pmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \begin{pmatrix} 1 & 2 & -1 \\ 0 & -1 & 3 \end{pmatrix} \xrightarrow{R_2 \to -R_2} \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & -3 \end{pmatrix}.$$

Then $R_1 \to R_1 - 2R_2$ gives

$$\begin{pmatrix} 1 & 0 & 5 \\ 0 & 1 & -3 \end{pmatrix}.$$

Therefore $x = -5z$ and $y = 3z$, so the solution set is

$$\left\{ z \begin{pmatrix} -5 \\ 3 \\ 1 \end{pmatrix} : z \in \mathbb{R} \right\}.$$

It is nonempty and closed under addition and scalar multiplication.

Theorem 2.5 (Homogeneous solution spaces)**.** Let $A \in M_{m,n}(\mathbb{F})$. The set

$$W = \{ x \in \mathbb{F}^n : Ax = \mathbf{0} \}$$

is a subspace of $\mathbb{F}^n$.

Proof. The zero vector is a solution. If $Ax = \mathbf{0}$ and $Ay = \mathbf{0}$, then

$$A(x+y) = Ax + Ay = \mathbf{0}, \qquad A(\alpha x) = \alpha(Ax) = \mathbf{0}.$$

The subspace test applies. □

If $b \neq \mathbf{0}$, the solution set of $Ax = b$ is never a subspace of $\mathbb{F}^n$, since $A\mathbf{0} = \mathbf{0} \neq b$. This remains true if the system has no solutions: an empty set is not a vector space.

Example 2.8 (Conservation and a physical constraint)**.** Suppose two inflows and one outflow satisfy $x_1 + x_2 - x_3 = 0$. The set of all real triples satisfying this equation is a subspace. For example,

$$\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ 4 \\ 5 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \\ 8 \end{pmatrix}$$

still satisfies the equation.

If we additionally require every flow to be nonnegative, the resulting set is not a subspace: multiplying the nonzero admissible vector $(2, 1, 3)^{\mathsf{T}}$ by -1 violates nonnegativity. A linear conservation equation and a nonnegativity constraint have different algebraic effects.

Example 2.9 (A binary parity-check space)**.** Over $\mathbb{Z}_2$, consider $x_1 + x_2 + x_3 = 0$. Its solution space is

$$W = \left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

For example, $(1, 1, 0)^{\mathsf{T}} + (1, 0, 1)^{\mathsf{T}} = (0, 1, 1)^{\mathsf{T}}$ because $1 + 1 = 0$ in $\mathbb{Z}_2$. The only scalars are 0 and 1, so scalar multiplication either gives zero or leaves the vector unchanged. This is a vector space over $\mathbb{Z}_2$, not over $\mathbb{R}$ with these same operations.

3 Intersections, unions, and sums of subspaces

3.1 Intersection and union

Example 3.1 (Two coordinate planes)**.** Let

$$U = \left\{ \begin{pmatrix} a \\ b \\ 0 \end{pmatrix} : a, b \in \mathbb{R} \right\}, \qquad W = \left\{ \begin{pmatrix} 0 \\ b \\ c \end{pmatrix} : b, c \in \mathbb{R} \right\}.$$

Their intersection is the common line

$$U \cap W = \left\{ \begin{pmatrix} 0 \\ b \\ 0 \end{pmatrix} : b \in \mathbb{R} \right\},$$

which is a subspace. Their union consists of vectors lying in at least one of the two planes. It is not closed under addition, since

$$\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \in U, \qquad \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix} \in W, \qquad \begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix} \notin U \cup W.$$

Theorem 3.1 (Intersection and union of two subspaces)**.** Let U, W be subspaces of a vector space V over any field $\mathbb{F}$.

1. $U \cap W$ is a subspace of V.
2. $U \cup W$ is a subspace if and only if $U \subseteq W$ or $W \subseteq U$.

More generally, the intersection of any family of subspaces is a subspace.

Proof. The zero vector belongs to every subspace, hence to their intersection. If two vectors belong to every member of the family, then so do their sum and every scalar multiple. The subspace

test proves the intersection assertion, even for an infinite family. With the usual convention, the intersection of an empty family of subsets of V is V itself.

For the union statement, if $U \subseteq W$, then $U \cup W = W$ is a subspace; the other containment case is identical. Conversely, suppose neither subspace contains the other. Choose

$$u \in U \setminus W, \qquad w \in W \setminus U.$$

If $u + w \in U$, then $w = (u + w) - u \in U$, a contradiction. If $u + w \in W$, then $u = (u + w) - w \in W$, also a contradiction. Thus $u, w \in U \cup W$ but $u + w \notin U \cup W$, so the union is not a subspace. This proof works over finite fields as well as infinite fields. □

Remark 3.2 (Why the theorem says two subspaces)**.** The two-subspace statement should not be extended carelessly to larger finite families over finite fields. For example, the three distinct lines

$$\operatorname{span}\{(1,0)^{\mathsf{T}}\}, \quad \operatorname{span}\{(0,1)^{\mathsf{T}}\}, \quad \operatorname{span}\{(1,1)^{\mathsf{T}}\}$$

cover $\mathbb{Z}_2^2$, although no one line contains the others. Here each span consists of zero and the displayed vector; we define span formally below.

3.2 The sum of two subspaces

Definition 3.3 (Sum of subspaces)**.** For subspaces $U, W \subseteq V$, define

$$U + W = \{u + w : u \in U,\ w \in W\}.$$

The same set-sum notation can be defined for arbitrary subsets, but the subspace conclusions below use the assumption that U and W are subspaces.

Example 3.2 (The two planes together generate the whole space)**.** For the coordinate planes above, every vector can be written as

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ z \end{pmatrix}.$$

Thus $U + W = \mathbb{R}^3$. The decomposition need not be unique: for example,

$$\begin{pmatrix} 3 \\ 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 3 \\ 2 \end{pmatrix}.$$

Each decomposition uses one vector from each plane.

Theorem 3.4 (The smallest subspace containing both)**.** If U, W are subspaces of V, then $U + W$ is a subspace of V. It contains both U and W, and every subspace containing both also contains $U + W$.

Proof. The zero vector equals $\mathbf{0} + \mathbf{0}$, so it belongs to $U + W$. For $u_1, u_2 \in U$, $w_1, w_2 \in W$, and $\alpha \in \mathbb{F}$,

$$(u_1 + w_1) + (u_2 + w_2) = (u_1 + u_2) + (w_1 + w_2) \in U + W,$$

$$\alpha(u_1 + w_1) = \alpha u_1 + \alpha w_1 \in U + W.$$

Thus the sum is a subspace. Every $u \in U$ equals $u + \mathbf{0}$, and every $w \in W$ equals $\mathbf{0} + w$, so both are contained in the sum. Finally, any subspace containing U and W contains $u + w$ for each such pair, hence contains all of $U + W$. □

Example 3.3 (A matrix sum). Inside $M_2(\mathbb{R})$, let

$$U = \left\{ \begin{pmatrix} a & 0 \\ 0 & -a \end{pmatrix} : a \in \mathbb{R} \right\}, \qquad W = \left\{ \begin{pmatrix} 0 & b \\ 2b & 0 \end{pmatrix} : b \in \mathbb{R} \right\}.$$

Then

$$U + W = \left\{ \begin{pmatrix} a & b \\ 2b & -a \end{pmatrix} : a, b \in \mathbb{R} \right\}.$$

For instance,

$$\begin{pmatrix} 3 & -1 \\ -2 & -3 \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 0 & -3 \end{pmatrix} + \begin{pmatrix} 0 & -1 \\ -2 & 0 \end{pmatrix}.$$

Example 3.4 (When one subspace is already inside the other). Let

$$U = \left\{ \begin{pmatrix} a & b \\ c & -a \end{pmatrix} : a, b, c \in \mathbb{R} \right\}, \qquad W = \left\{ \begin{pmatrix} 0 & d \\ -d & 0 \end{pmatrix} : d \in \mathbb{R} \right\}.$$

Since $W \subseteq U$, their sum is U itself. Indeed,

$$\begin{pmatrix} a & b \\ c & -a \end{pmatrix} + \begin{pmatrix} 0 & d \\ -d & 0 \end{pmatrix} = \begin{pmatrix} a & b+d \\ c-d & -a \end{pmatrix} \in U.$$

The reverse containment follows by taking the second summand to be zero. More generally, $W \subseteq U$ always implies $U + W = U$.

Example 3.5 (A sum of polynomial subspaces). In $V = \mathbb{R}_3[x]$, let

$$U = \{ f \in V : f(2) = 0 \}, \qquad W = \{ f \in V : f(0) = 0 \}.$$

By the factor theorem, the polynomials in U have the form

$$(x-2)(ax^2 + bx + c) = ax^3 + (b-2a)x^2 + (c-2b)x - 2c.$$

The polynomials in W have the form $dx^3 + ex^2 + gx$. Hence

$$U + W = \left\{ (a+d)x^3 + (b-2a+e)x^2 + (c-2b+g)x - 2c : a, b, c, d, e, g \in \mathbb{R} \right\}.$$

In fact, $U + W = V$. To see this directly, let $f \in V$ and set

$$u(x) = f(0)\left(1 - \frac{x}{2}\right), \qquad w(x) = f(x) - u(x).$$

Then $u(2) = 0$, so $u \in U$, and $w(0) = f(0) - f(0) = 0$, so $w \in W$. Thus $f = u + w$. The decomposition is not unique: the nonzero polynomial $x(x-2)$ belongs to both subspaces.

3.3 Direct sums and uniqueness

Definition 3.5 (Direct sum). The sum $U + W$ is a *direct sum*, denoted $U \oplus W$, if every $v \in U + W$ has a unique representation $v = u + w$ with $u \in U$ and $w \in W$. Equivalently,

$$u_1 + w_1 = u_2 + w_2 \quad \Longrightarrow \quad u_1 = u_2 \text{ and } w_1 = w_2.$$

Theorem 3.6 (Direct-sum test). For subspaces U, W of V, their sum is direct if and only if

$$U \cap W = \{\mathbf{0}\}.$$

Proof. Suppose the sum is direct and $v \in U \cap W$. Then

$$v = v + \mathbf{0} = \mathbf{0} + v$$

are two decompositions using elements of U and W. Uniqueness forces $v = \mathbf{0}$.

Conversely, suppose $U \cap W = \{\mathbf{0}\}$ and $u_1 + w_1 = u_2 + w_2$. Rearranging gives

$$u_1 - u_2 = w_2 - w_1.$$

The left side belongs to U and the right side to W, so this common vector belongs to $U \cap W$ and is zero. Hence $u_1 = u_2$ and $w_1 = w_2$. □

Example 3.6 (Symmetric and skew-symmetric parts)**.** In $M_n(\mathbb{C})$, let U be the subspace of symmetric matrices and W the subspace of skew-symmetric matrices. Since $1 + 1 \neq 0$ in $\mathbb{C}$, a matrix that is both symmetric and skew-symmetric is zero. Therefore $U \cap W = \{0\}$ and the sum is direct.

Every $A \in M_n(\mathbb{C})$ also satisfies

$$A = \underbrace{\frac{1}{2}(A + A^{\mathsf{T}})}_{S} + \underbrace{\frac{1}{2}(A - A^{\mathsf{T}})}_{K}.$$

Indeed, $S^{\mathsf{T}} = S$ and $K^{\mathsf{T}} = -K$, so $S \in U$ and $K \in W$. Thus

$$M_n(\mathbb{C}) = U \oplus W.$$

For a concrete example,

$$\begin{pmatrix} 3 & 4 \\ -2 & 5 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix} + \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}.$$

The first matrix is symmetric and the second is skew-symmetric. The decomposition is unique. The same result holds over any field with $1 + 1 \neq 0$; division by two is essential in this argument.

4 Linear combinations and span

4.1 Linear combinations

Definition 4.1 (Linear combination)**.** Let $v_1, \ldots, v_n \in V$, where V is a vector space over $\mathbb{F}$. A *linear combination* of these vectors is a vector of the form

$$\alpha_1 v_1 + \cdots + \alpha_n v_n, \qquad \alpha_1, \ldots, \alpha_n \in \mathbb{F}.$$

The scalars α_j are called the coefficients of the linear combination.

Example 4.1 (From a line to a plane to the whole space)**.** In $\mathbb{R}^3$, take

$$v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad v_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad v_3 = \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}.$$

The combinations of v_1 alone are $(\alpha, 0, 0)^{\mathsf{T}}$, forming a line. The combinations of v_1, v_2 are

$$\alpha_1 v_1 + \alpha_2 v_2 = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ 0 \end{pmatrix},$$

forming the plane with third coordinate zero. With all three vectors, the combinations are

$$\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 = \begin{pmatrix} \alpha_1 \\ \alpha_2 + 2\alpha_3 \\ \alpha_3 \end{pmatrix}.$$

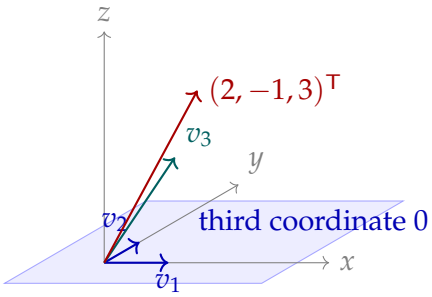
For example,

$$2v_1 - 7v_2 + 3v_3 = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}.$$

This target cannot be obtained from v_1, v_2 alone because its third coordinate is nonzero. In general, for any $(x, y, z)^{\mathsf{T}} \in \mathbb{R}^3$, choose

$$\alpha_1 = x, \qquad \alpha_2 = y - 2z, \qquad \alpha_3 = z.$$

Then the displayed combination gives the target vector. We say that v_1, v_2, v_3 *span* $\mathbb{R}^3$.



Theorem 4.2 (Subspaces contain linear combinations)**.** If W is a subspace of V and $w_1, \ldots, w_n \in W$, then every linear combination of these vectors belongs to W.

Proof. For one vector this is closure under scalar multiplication. Suppose the claim is true for $n-1$ vectors. The sum $\alpha_1 w_1 + \cdots + \alpha_{n-1} w_{n-1}$ belongs to W by the induction hypothesis, and $\alpha_n w_n \in W$ by scalar closure. Closure under addition proves the statement for n vectors. □

By convention, the linear combination of an empty collection is the zero vector.

4.2 Testing whether a linear combination exists

Example 4.2 (A membership question becomes a linear system)**.** Let

$$v = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \qquad w_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \qquad w_2 = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}.$$

To ask whether v is a linear combination of w_1, w_2 is to ask whether there are real α, β satisfying

$$\alpha w_1 + \beta w_2 = v.$$

Entry by entry, the conditions are

$$\begin{cases} \alpha + \beta = 1, \\ \alpha - \beta = 1, \\ \alpha + 2\beta = 0. \end{cases}$$

We reduce the augmented matrix:

$$\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 2 & 0 \end{array}\right) \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - R_1} \left(\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & -2 & 0 \\ 0 & 1 & -1 \end{array}\right).$$

Next, $R_2 \to -\frac{1}{2}R_2$ and then $R_3 \to R_3 - R_2$ give

$$\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{array}\right).$$

The last row says $0 = -1$. Hence the system is inconsistent, and v is not a linear combination of the two given vectors. This is a general method: place the proposed generating vectors in columns and row-reduce the augmented matrix with the target as its last column.

Theorem 4.3 (A matrix equation and its columns)**.** Let $A \in M_{m,n}(\mathbb{F})$ and $b \in \mathbb{F}^m$. The system $Ax = b$ is solvable if and only if b is a linear combination of the columns of A.

Proof. Write the columns as $a_1, \dots, a_n$. For $x = (x_1, \dots, x_n)^{\mathsf{T}}$, matrix multiplication gives

$$Ax = x_1 a_1 + \cdots + x_n a_n,$$

because the ith entry on either side is

$$a_{i1} \times x_1 + \cdots + a_{in} \times x_n.$$

Thus a solution vector supplies exactly the coefficients of the required linear combination, and conversely any such coefficients supply a solution. □

Example 4.3 (Combining two command patterns)**.** Suppose a three-component command is built from the patterns

$$p = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \qquad q = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

A combination has the form

$$\alpha p + \beta q = \begin{pmatrix} \alpha \\ \beta \\ \alpha + \beta \end{pmatrix}.$$

Thus $(2, 3, 5)^{\mathsf{T}} = 2p + 3q$ is attainable. The target $(2, 3, 4)^{\mathsf{T}}$ is not: its first two coordinates force $\alpha = 2$ and $\beta = 3$, giving third coordinate 5. The attainable commands form the plane $z = x + y$ through the origin.

4.3 Span and spanning sets

Definition 4.4 (Span)**.** For $v_1, \dots, v_n \in V$, define

$$\operatorname{span}\{v_1, \dots, v_n\} = \{\alpha_1 v_1 + \cdots + \alpha_n v_n : \alpha_1, \dots, \alpha_n \in \mathbb{F}\}.$$

This is the set of all linear combinations of the listed vectors. By convention,

$$\operatorname{span} \varnothing = \{\mathbf{0}\}.$$

For an arbitrary subset $S \subseteq V$, its span is the set of all linear combinations of *finitely many* elements of S.

Remark 4.5. The listed vectors may be repeated, and their order may be changed without changing the span. Even if S is infinite, every combination in its span is finite. No infinite sums are being introduced here. The field also matters: coefficients must belong to the specified scalar field.

Example 4.4 (Coordinate vectors and polynomials)**.** For the standard coordinate vectors in $\mathbb{R}^3$,

$$\operatorname{span}\left\{\begin{pmatrix}1\\0\\0\end{pmatrix},\begin{pmatrix}0\\1\\0\end{pmatrix},\begin{pmatrix}0\\0\\1\end{pmatrix}\right\}=\mathbb{R}^3.$$

Likewise,

$$\operatorname{span}\{1,x,x^2,x^3,x^4\}=\mathbb{F}_4[x]$$

over $\mathbb{F}$, since an arbitrary polynomial of degree at most four is a linear combination of these monomials.

Example 4.5 (The earlier matrix space as a span)**.** Let

$$A=\begin{pmatrix}1&0\\0&0\\2&1\end{pmatrix},\qquad B=\begin{pmatrix}0&1\\-2&0\\-1&4\end{pmatrix}.$$

Then, over $\mathbb{C}$,

$$\operatorname{span}\{A,B\}=\left\{\begin{pmatrix}a&b\\-2b&0\\2a-b&a+4b\end{pmatrix}:a,b\in\mathbb{C}\right\}.$$

Indeed, the displayed matrix is exactly $aA+bB$, so this is the subspace W from an earlier example. In particular,

$$\begin{pmatrix}1&-1\\2&0\\3&-3\end{pmatrix}=A-B$$

belongs to the span. Changing its middle-right entry from 0 to 1 produces a matrix outside the span, since every combination has zero in that position. With real coefficients instead, the same two real matrices span the corresponding real subspace.

Definition 4.6 (Spanning set)**.** A list $v_1,\dots,v_n$ *spans* V if

$$\operatorname{span}\{v_1,\dots,v_n\}=V.$$

Equivalently, for every $v\in V$, there are coefficients $a_1,\dots,a_n\in\mathbb{F}$ such that $v=a_1v_1+\cdots+a_nv_n$.

Example 4.6 (A finite list and an infinite space)**.** The polynomials $1,x,x^2,x^3$ span $\mathbb{R}_3[x]$. No finite list spans $\mathbb{R}[x]$: for any finite list of polynomials, choose a degree bound d for all of them. Every linear combination has degree at most d or is zero, so the polynomial x^{d+1} is outside their span.

The scalar field changes what a spanning list must do. The list $1,i$ spans $\mathbb{C}$ over $\mathbb{R}$, since $a+ib=a\,1+b\,i$ with real a,b. Over $\mathbb{C}$, the single vector 1 already spans $\mathbb{C}$.

Definition 4.7 (Finite-dimensional vector space)**.** A vector space is *finite-dimensional* if it has a finite spanning set. The space $\mathbb{R}[x]$ is an example that is not finite-dimensional.

Theorem 4.8 (Span is the smallest containing subspace)**.** For vectors $v_1,\dots,v_n\in V$, the set $\operatorname{span}\{v_1,\dots,v_n\}$ is a subspace of V containing all the listed vectors. Every subspace containing those vectors contains their span.

The same statement holds for the span of any subset $S\subseteq V$.

Proof. Taking all coefficients to be zero puts $\mathbf{0}$ in the span. Adding two linear combinations adds their corresponding coefficients, and multiplying a combination by a scalar multiplies each coefficient by that scalar. The result is again a linear combination, so the subspace test applies. Each v_j belongs to the span by taking its coefficient to be 1 and all others to be 0.

Any subspace containing the listed vectors contains every linear combination of them, by the earlier closure theorem, so it contains their span. For a general subset S, the same proof uses the finite collections of vectors appearing in the combinations. In the empty case the span is $\{\mathbf{0}\}$, which is contained in every subspace. $\square$